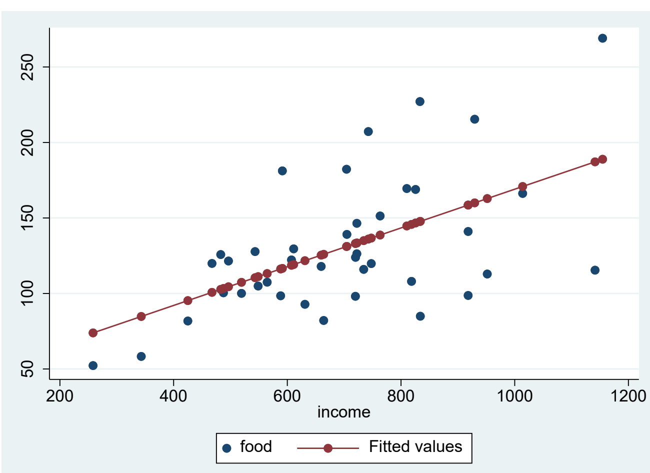
```
. /* Demo program for the tutorial */
. /* To download the files into your harddisk, point your mouse
>    to the file, click on the right button and choose "Save
>    target as ...".
>    Suppose you have downloaded the files food.txt and food.dta
>    in the directory \simpson\homer on your harddisk.
>    If you want to change Stata to the directory, type in the
>    command window:
>    cd "\simpson\homer"
> */
.
. clear    /* clear everything in the memory. A good practice */
.
. dir /* Show the contents in the current directory. If you do not see the
>        two data files, issue the following commands to change directory */
  <dir>   4/20/21 19:05  .
  <dir>   4/20/21 19:05  ..
   2.9k   8/29/19 14:11  food.do
  74.7k   8/29/19 14:13  food.docx
   0.7k   1/31/02 22:52  food.dta
   0.2k   4/20/21 19:05  food.log
 119.4k   8/29/19 14:14  food.pdf
   0.5k   1/29/02 18:34  food.txt
 116.1k   8/14/20 16:39  food.zip
   0.9k   1/10/12 15:58  food_data.zip
.
. // cd \simpson\homer     /* change directory if necessary */
.
. /* Two ways to read data */
.
. /* Method 1: If you have a text file with variable names on the first row */
.
. insheet using food.txt  /* Now you should see two variables in the Variable
>                            Window. To save it in Stata format, try the
>                            command "save myfood, replace"                 */
(2 vars, 40 obs)

.
. /* Method 2: If you have a data file in Stata format already */
. use food, clear      /* Try "use myfood, clear" to read the file
>                         you just saved.                        */

.
. d         /* describe the data set and variables */

Contains data from food.dta
  obs:            40
 vars:             2                          31 Jan 2002 21:52
--------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
```

```
--------------------------------------------------------------------------------
food            float   %9.0g
income          float   %9.0g
--------------------------------------------------------------------------------
Sorted by:

. sum       /* summary statistics                 */

    Variable |       Obs        Mean    Std. Dev.       Min        Max
-------------+-----------------------------------------------------
        food |        40     130.313    45.15857      52.25     269.03
      income |        40         698    198.2269      258.3     1154.6

. insp      /* inspect the data                   */

food:                                         Number of Observations
-------                                  ---------------------------------------
                                              Total      Integers   Nonintegers
|      #                      Negative            -             -             -
|      #                      Zero                -             -             -
|      #                      Positive           40             -            40
|      #                                  -----------   -----------   -----------
|      #   #                  Total              40             -            40
|  #   #   #   .   .          Missing             -
+----------------------                   -----------
52.25            269.03                          40
  (40 unique values)

income:                                       Number of Observations
---------                                ---------------------------------------
                                              Total      Integers   Nonintegers
|          #                  Negative            -             -             -
|      #   #                  Zero                -             -             -
|      #   #   #              Positive           40             6            34
|      #   #   #                          -----------   -----------   -----------
|      #   #   #              Total              40             6            34
|  #   #   #   #   #          Missing             -
+----------------------                   -----------
258.3            1154.6                          40
  (38 unique values)

.
. * Generate a new variable
. generate income2 = income^2

.
. reg food income   /* regression          */

      Source |       SS           df       MS      Number of obs   =        40
-------------+----------------------------------   F(1, 38)        =     17.65
       Model |   25221.219         1   25221.219   Prob > F        =    0.0002
```

```
    Residual |  54311.3331        38  1429.24561   R-squared       =    0.3171
-------------+----------------------------------   Adj R-squared   =    0.2991
       Total |  79532.5521        39  2039.29621   Root MSE        =    37.805

------------------------------------------------------------------------------
        food |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      income |   .1282886   .0305393     4.20   0.000     .0664651    .1901121
       _cons |   40.76756   22.13865     1.84   0.073    -4.049799    85.58493
------------------------------------------------------------------------------

. test income=0     /* hypothesis testing  */

 ( 1)  income = 0

       F(  1,    38) =   17.65
            Prob > F =    0.0002

. test income=1

 ( 1)  income = 1

       F(  1,    38) =  814.76
            Prob > F =    0.0000

.
. predict foodhat   /* prediction          */
(option xb assumed; fitted values)

. twoway (scatter food income) (connected foodhat income) /* graph */
```

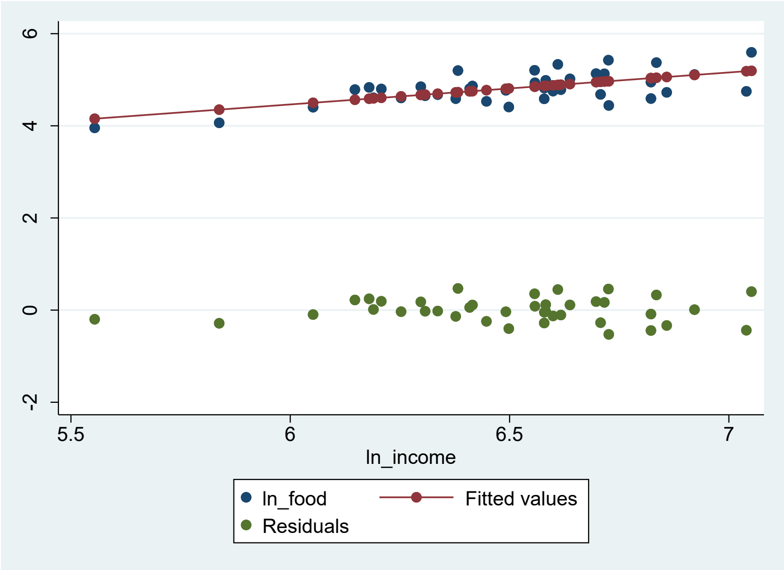



```
.
. * Generate residuals both ways and see if they are the same.
. predict ehat, resid

. gen ehat_my = food - _b[_cons] - _b[income]*income
```

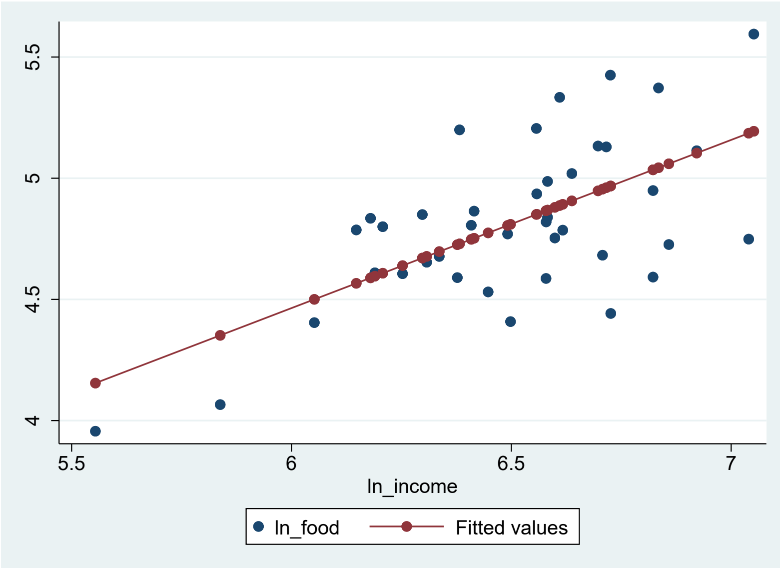
```
. * Use Data Browse to see if they are the same.
.
. * Only use some of the observations in a regression or in a graph
. regress food income if income <720

      Source |       SS           df       MS      Number of obs   =        21
-------------+----------------------------------   F(1, 19)        =      9.10
       Model |  6751.86607         1  6751.86607   Prob > F        =    0.0071
    Residual |  14094.3355        19  741.807133   R-squared       =    0.3239
-------------+----------------------------------   Adj R-squared   =    0.2883
       Total |  20846.2016        20  1042.31008   Root MSE        =    27.236

------------------------------------------------------------------------------
        food |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      income |   .1550401   .0513899     3.02   0.007     .0474798    .2626004
       _cons |   25.85125   29.04663     0.89   0.385    -34.94405    86.64655
------------------------------------------------------------------------------

.
. /* Practice Question: Consider a log-log model.
> 1. Read the data (either the Stata format or the text format) into Stata.
> 2. Create two new variables ln_food = ln(food), ln_income = ln(income).
> 3. Use describe, summarize, and inspect commands on the new data.
> 4. Regress ln_food on ln_income. How do you interpret the slope coeff?
> 5. Create yhat and ehat for the new model.
> 6. Draw a graph with ln_income on the horizontal axis, the actual ln_food,
>    yhat, and ehat on the vertical axis.
> */
. * See next page for answers
```

```
. use food, clear

. generate ln_food = ln(food)

. generate ln_income = ln(income)

. describe

Contains data from food.dta
  obs:            40
 vars:             4                         31 Jan 2002 21:52
-------------------------------------------------------------------------------
              storage  display    value
variable name   type   format     label      variable label
-------------------------------------------------------------------------------
food            float  %9.0g
income          float  %9.0g
ln_food         float  %9.0g
ln_income       float  %9.0g
-------------------------------------------------------------------------------
Sorted by:
     Note: Dataset has changed since last saved.

. summarize ln_income ln_food

    Variable |       Obs        Mean    Std. Dev.       Min        Max
-------------+--------------------------------------------------------
   ln_income |        40    6.505224    .3077564   5.554121   7.051509
     ln_food |        40     4.81468     .337136    3.95604   5.594823

. inspect ln_income ln_food

ln_income:                                      Number of Observations
------------                            ---------------------------------------
                                              Total      Integers   Nonintegers
|              #              Negative            -             -             -
|              #              Zero                -             -             -
|          #   #              Positive           40             -            40
|          #   #                        -----------   -----------   -----------
|          #   #   #          Total              40             -            40
|  .   .   #   #   #          Missing             -
+----------------------                 -----------
5.554121       7.051509                          40
  (38 unique values)

ln_food:                                        Number of Observations
----------                              ---------------------------------------
                                              Total      Integers   Nonintegers
|          #                  Negative            -             -             -
|          #                  Zero                -             -             -
|          #                  Positive           40             -            40
```

```
|      #    #                                 -----------   -----------   -----------
|      #    #    #                 Total               40             -            40
|  .   #    #    #    #            Missing              -
+----------------------                                 -----------
3.95604          5.594823                                       40
  (40 unique values)

. regress ln_food ln_income

      Source |       SS           df       MS      Number of obs   =        40
-------------+----------------------------------   F(1, 38)        =     25.48
       Model |  1.77932063         1  1.77932063   Prob > F        =    0.0000
    Residual |  2.65344666        38  .069827544   R-squared       =    0.4014
-------------+----------------------------------   Adj R-squared   =    0.3856
       Total |  4.43276729        39    .1136607   Root MSE        =    .26425

------------------------------------------------------------------------------
     ln_food |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
   ln_income |   .6940451   .1374909     5.05   0.000     .4157093    .9723808
       _cons |   .2997616   .8953845     0.33   0.740    -1.512849    2.112373
------------------------------------------------------------------------------

. predict yhat
(option xb assumed; fitted values)

. predict ehat, residuals

. gen ehat_my = ln_food - _b[_cons] - _b[ln_income]*ln_income

. // The first graph scale is off because of the residuals; it does not look
nice
. twoway (scatter ln_food ln_income) (connected yhat ln_income) (scatter ehat
ln_income)
```



```
. twoway (scatter ln_food ln_income) (connected yhat ln_income)
```




```
. twoway (scatter ehat ln_income)
```